

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Python**-based applications and research tooling, delivering key components for quantitative researchers' analytical workflows.
- Engineered and enhanced large-scale **datasets** used across research and trading teams, focusing on data integrity and efficient querying.
- Designed and implemented **data-driven analytical workflows** and model features, directly supporting the generation of actionable insights for trading strategies.
- Optimised algorithms and improved the performance of existing **Python** code, reducing execution times by an average of **30%** for critical data processing tasks.
- Collaborated directly with **quantitative researchers** and traders on high-impact projects, translating complex data requirements into robust engineering solutions.
- Managed **production environments** on Linux, ensuring stability and performance for data-intensive applications supporting research activities.
- Containerised data processing applications using **Docker** and deployed them to **AWS EC2** instances, enhancing deployment consistency and scalability.

SKILLS

Languages – Python, SQL, Java

Libraries & Frameworks – Pandas, NumPy, Scikit-learn, Spring Boot, FastAPI, Flask

Data Platforms & Databases – PostgreSQL, MySQL, Oracle, AWS S3, AWS Lambda

Tools & Technologies – Linux, Git, Jenkins, Docker, AWS EC2, Splunk, Dynatrace

Methodologies & Concepts – Data Engineering, Algorithmic Thinking, Agile, REST APIs, CI/CD, Data Warehousing, Machine Learning Fundamentals

PROJECTS

Quantitative Data Pipeline for Trading Research

- Developed a **Python** data pipeline using **Pandas** and **SQL** to ingest and transform large volumes of market data, supporting a **quantitative research** team.
- Implemented data validation checks and error handling to ensure the integrity of the processed data, reducing data-related issues in downstream analyses by **40%**.
- Built modular components for feature extraction and aggregation, enabling researchers to quickly test new hypotheses and trading strategies.

Algorithmic Performance Optimisation Service

- Engineered a **Python** service to identify and optimise inefficient algorithms in existing research code, applying principles of algorithmic complexity analysis.
- Refactored critical data processing routines, achieving performance improvements that reduced average computation time by approximately **28%**.
- Integrated the optimisation service into the research team's workflow, providing automated performance profiling and actionable recommendations for code enhancement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)